MATHEMATICS BOARDS DOMINATOR: LAST WEEK GUIDE

By Aashi Mathematics • Targeted Syllabus Analysis (2026)

STEP 0 — INPUT CHECK

Analysis of Series 65-5-1, 65-7-1 marks this as the **2023-24 Case Study Pattern**. These sets align with the **New Reduced Syllabus**.

STEP 1 — SYLLABUS FILTERING

VALID: Matrices (AX=B), Calculus (King's Rule, Area of Ellipse), 3D Lines (Shortest Distance), Bayes’ Theorem.
EXCLUDED: Planes (3D), Binomial Distribution, and Elementary Row Operations are NOT appearing.

STEP 2 — CHAPTER WEIGHTAGE

Chapter	Importance	Trend Info
Calculus	🔥 **HIGHEST**	12-14 Qs. Focus on Integrals & DE.
Vectors & 3D	**High**	Shortest Distance is a fixed 5-marker.
Matrices	**High**	Solving AX=B is guaranteed Section D.
Probability	**Case Study**	Bayes' Theorem is the standard for 4-marks.

STEP 3 — TOP REPEATED TOPICS

1. Matrices (AX=B) | 2. 3D Shortest Distance | 3. LPP Graphical | 4. Area Under Curve | 5. Bayes' Theorem | 6. Linear DE (I.F.) | 7. Equivalence Relations | 8. Dot/Cross Vectors | 9. |adj A| Properties | 10. Maxima/Minima Cases.

STEP 5 — PREDICTION ZONE

🔥 **MUST PREPARE (Sure-Shot):** Matrices (Solving 3 variables), 3D Lines (SD Formula), LPP Max/Min, Bayes' Case Study.
⚡ **HIGH CHANCE:** Linear Diff Eq, Area of Ellipse/Circle, Equivalence proofs.

STEP 6 — PRACTICE QUESTION EXCERPTS (SELECTED FROM 40 QS SET)

Matrices: Solve x-y+2z=7, 3x+4y-5z=-5, 2x-y+3z=12.

3D: Find SD between r = (i+2j+k)+λ(i-j+k) and r = (2i-j-k) +μ(2i+j+2k).

Integrals: Evaluate $\int_0^{\alpha/2}$ (√sin x)/(√sin x + √cos x) dx. (Ans: π/4)

Probability: Bayes' problem on defective items from two bags.

Diff Eq: Solve dy/dx + y cot x = 2x + x^2 cot x.

Vectors: Find λ if (2i+3j-k) and (i+λj+2k) are perpendicular.

STEP 7 — QUICK REVISION & TRAPS

1-Page Revision: I.F. = e^∫P dx | |kA| = k^n|A| | r = a + λb (Line Eq) | P(E|A) Bayes.
Traps: Area can't be negative (use modulus) | Don't forget +C in Integrals | Coefficient of x,y,z must be 1 in 3D.

STEP 8 — HONEST REPORT

The pattern is stable across 4 sets. 3D Geometry is now purely **Line-based**. Do not waste time on Planes or Angle between Planes.